

C6.2 Organic chemistry

Summary

Carbon chemistry is the basis of life on Earth. Organic chemistry is the basis of many of the materials we produce. Organic compounds are covalent in nature and react in a predictable pattern. Crude oil forms the basis of many useful by-products.

Underlying knowledge and understanding

Learners should be familiar with reactions and displayed formula.

Common misconceptions

Learners tend not to bring the concepts from general chemistry in their study of organic chemistry. They have difficulty identifying functional groups and naming and drawing the compounds.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
CM6.2i ☒	represent three-dimensional shapes in two dimensions and vice versa when looking at chemical structures, e.g. allotropes of carbon	M5b

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
C6.2a ☒	recognise functional groups and identify members of the same homologous series			
C6.2b ☒	name and draw the structural formulae, using fully displayed formulae, of the first four members of the straight chain alkanes, alkenes, alcohols and carboxylic acids	M5b	WS1.4a	Use of models.
C6.2c ☒	predict the formulae and structures of products of reactions of the first four and other given members of the homologous series of alkanes, alkenes and alcohols			
C6.2d ☒	recall the basic principles of addition polymerisation by reference to the functional group in the monomer and the repeating units in the polymer			
C6.2e ☒	explain the basic principles of condensation polymerisation		WS1.4a	
	reference to the functional groups of the monomers, the minimum number of functional groups within a monomer, the number of repeating units in the polymer, and simultaneous formation of a small molecule, e.g. a polyester or polyamide, using block diagrams to represent polymers			

C6.2f ☒	describe practical techniques to make a polymer by condensation			WS1.2a, WS1.2b, WS1.2c, WS1.4a, WS2a, WS2b	Making nylon.
C6.2g ☒	deduce the structure of an addition polymer from a simple alkene monomer and vice versa	the following representation of a polymer [repeat unit] _n		WS1.4a	
C6.2h ☒	recall that DNA is a polymer made from four different monomers called nucleotides and that other important naturally-occurring polymers are based on sugars and amino-acids	the names of the nucleotides		WS1.4a	
C6.2i ☒	recall that it is the generality of reactions of functional groups that determine the reactions of organic compounds			WS1.4a	
C6.2j	describe the separation of crude oil by fractional distillation	the names of the fractions		WS1.3f, WS1.4a	
C6.2k	explain the separation of crude oil by fractional distillation	molecular size and intermolecular forces			
C6.2l	describe the fractions as largely a mixture of compounds of formula C _n H _{2n+2} which are members of the alkane homologous series			WS1.4a	
C6.2m	recall that crude oil is a main source of hydrocarbons and is a feedstock for the petrochemical industry			WS1.4a	
C6.2n	explain how modern life is crucially dependent upon hydrocarbons and recognise that crude oil is a finite resource			WS1.1c, WS1.1f, WS1.1e, WS1.4a	

C6.2o	describe the production of materials that are more useful by cracking	conditions and reasons for cracking and some of the useful materials produced			
C6.2p ☒	recall that a chemical cell produces a potential difference until the reactants are used up				
C6.2q ☒	evaluate the advantages and disadvantages of hydrogen/oxygen and other fuel cells for given uses	the chemistry of the hydrogen/oxygen fuel cell		WS1.1g, WS1.1i	